Swiss Institute of Bioinformatics

DAY 2, PART 3

AlphaFold Server tutorial

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23-24 June 2026

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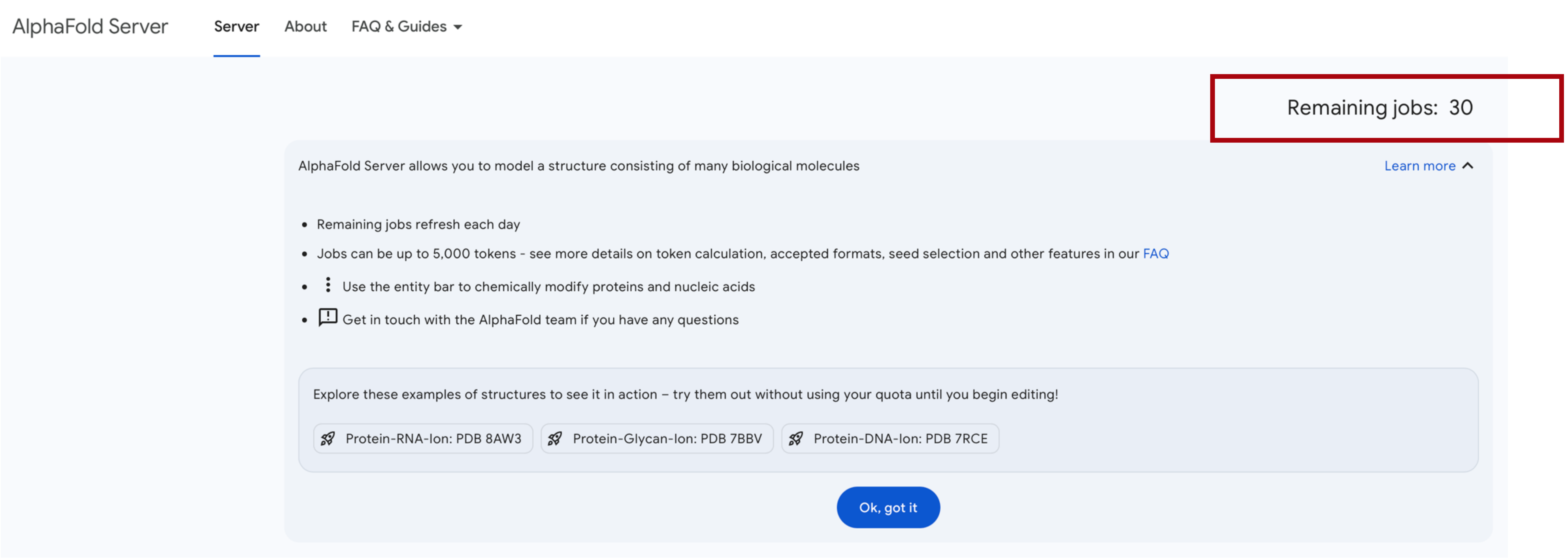
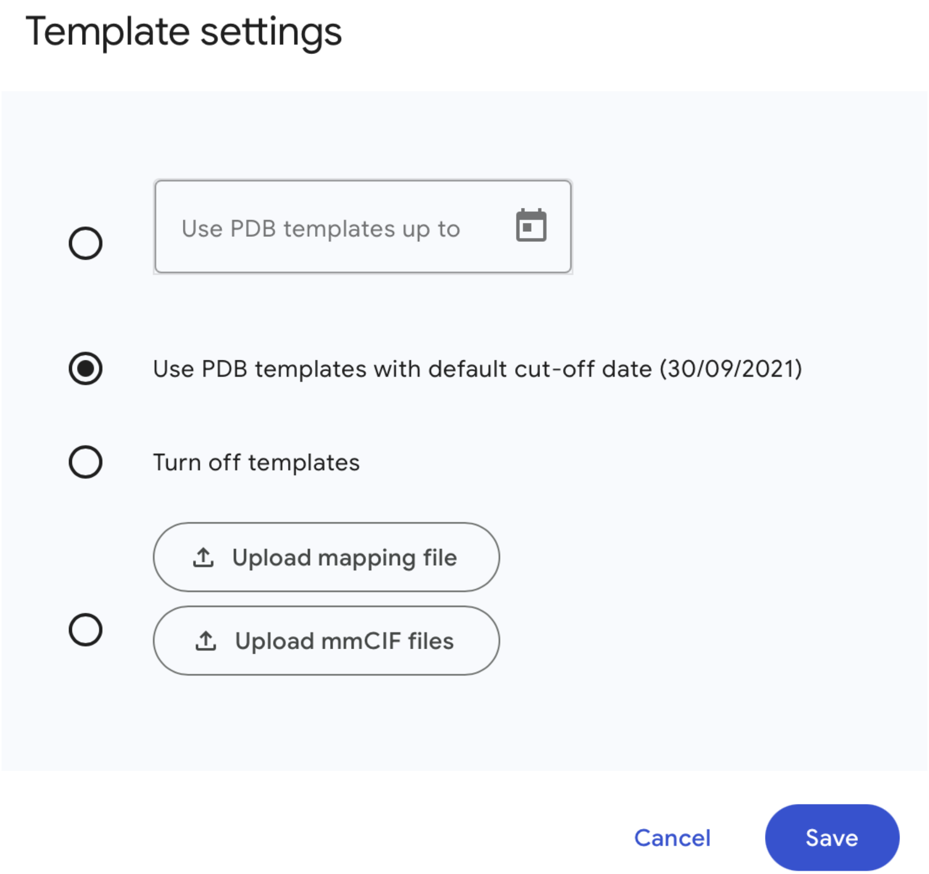
AlphaFold Server

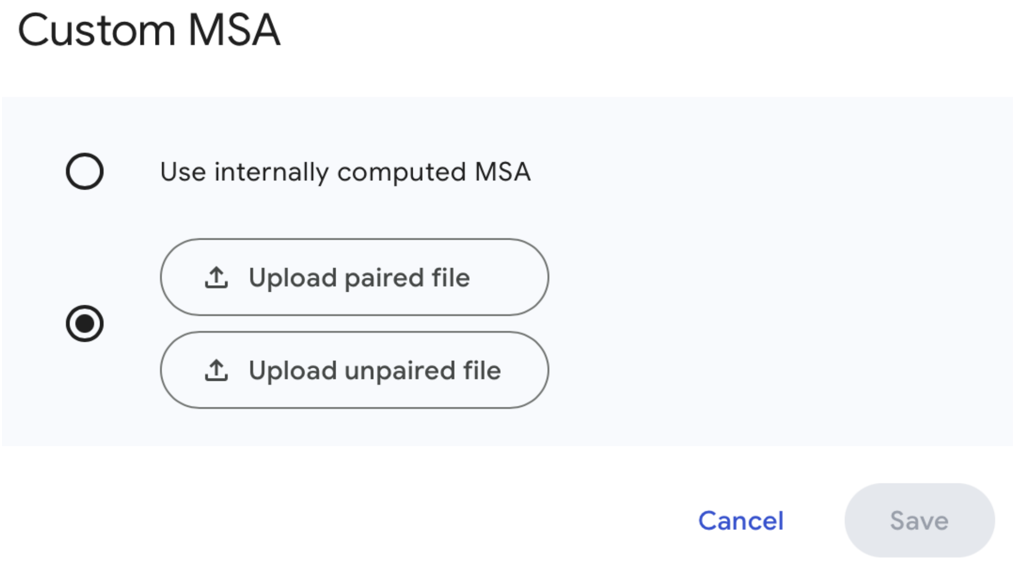
- Free and you do not need to install any software on your computer
- Supports modeling of multi-entity complexes (proteins, DNA, RNA, ligands, ions), but exposes only a limited subset of the full AlphaFold 3 (check FAQs)
- The minimum sequence length is four amino acids or four nucleotides
- Each modelling job is limited to 5,000 tokens
- Each user is limited to 30 jobs per day



AlphaFold Server: Settings

- There are options for custom templates and MSA
- You can specify a cut-off date for the PDB templates to be used. The cut-off date for the template is usually used if you don't want templates that AF3 was trained on (training cut-off date: 2021-09-30).



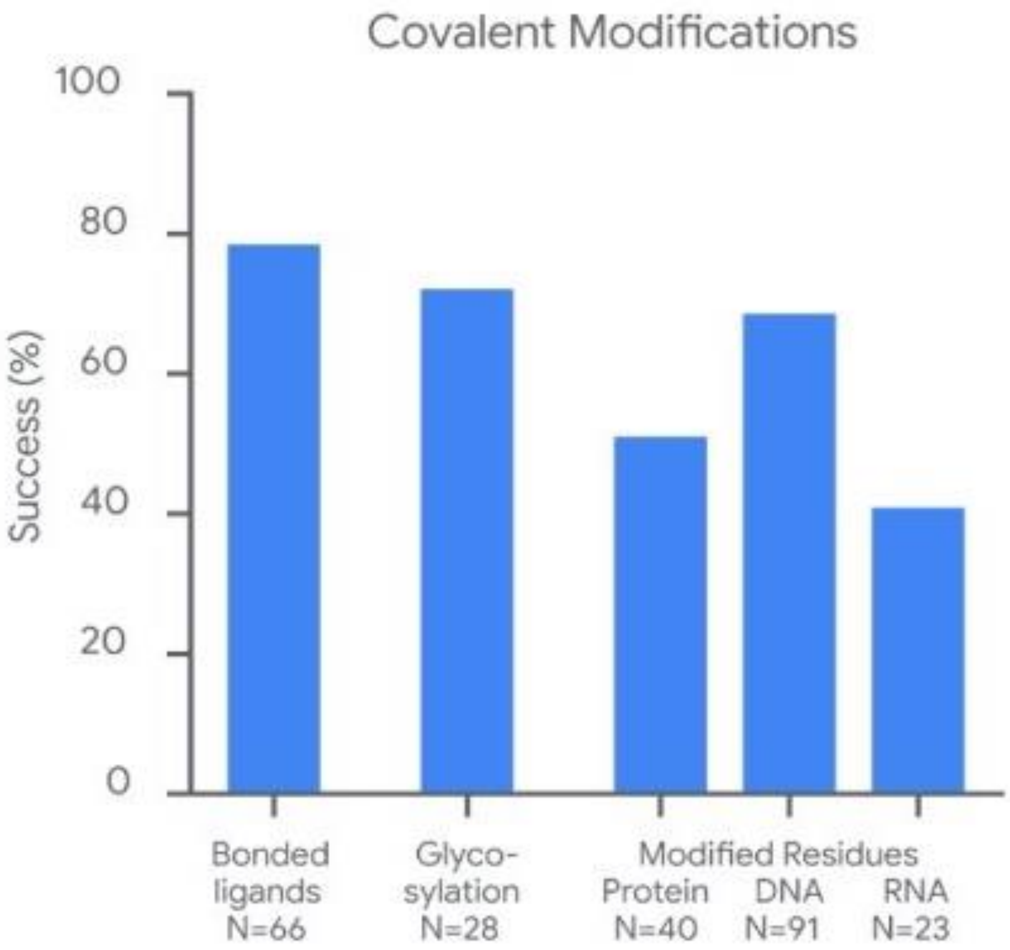


AlphaFold Server: Settings

Adding modifications

- Multiple modifications to different residues one at a time (for Glycosylated proteins see AlphaFold Server FAQ)

- Modifications for DNA and RNA



Abramson et al, 2024

AlphaFold Server: Advanced Settings

- **JSON job submission** for automatic job generation (useful for screening experiments, up to 100 jobs per file; you are allowed up to 500 saved drafts)

- **Sampling multiple seeds** (20 seeds is usually enough to get a better prediction)

- **Reproducing jobs** (just use the same input and seed number, which can be found in fold_<job_name>_job_request.json - contains all the information required for reproduction)

AlphaFold Server: Output

- Five .cif files named **fold_<job_name>_model_<N>cif** contain predicted structures in the mmCIF format (can be viewed in any molecular viewer);
- Five .json files named **fold_<job_name>_summary_confidences_<N>.json** contain summaries of the confidence metrics for the predictions;
- Five .json files named **fold_<job_name>_full_data_<N>.json** contain detailed confidence metrics (per-atom pLDDT, per-token contact prob matrix, per-token pae etc.)
- A file named **fold_<job_name>_job_request.json** contains the inputs of the modelling job and could be used to re-run the job;
- A file named **terms_of_use.md**. This is a legal document detailing the terms of use for the predictions.

where N is the rank of the predicted structure, structures are ranked from 0 to 4, where 0 has the highest confidence

AlphaFold Server: Confidence metrics

JSON files with **summary** contain:

- **chain_iptm:** the average confidence (ipTM) in the interfaces between each chain and all other chains, can be used for ranking predicted structures for a specific chain;
- **chain_pair_iptm:** pairwise ipTM scores, can be used for ranking predictions of a structure by the accuracy of a specific interface between two chains;
- **chain_pair_pae_min:** element (i, j) of the array contains the lowest PAE value across rows restricted to chain i and columns restricted to chain j. This has been found to correlate with whether or not two chains interact;
- **chain_ptm:** pTM restricted to chain i;
- **fraction_disordered:** range 0-1 that indicates what fraction of the prediction structure is disordered;
- **has_clash:** 1.0 if more than 50% of a chain's atoms clash, or if a chain has more than 100 clashing atoms; 0.0 otherwise.
- **iptm:** ipTM for all interfaces in the structure;
- **ptm:** pTM-score for the full structure;
- **ranking_score:** 0.8 × ipTM + 0.2 × pTM + 0.5 × disorder – 100 × has_clash, range from –100 to 1.5

AlphaFold Server: Confidence metrics

JSON files with **full outputs** contain:

- **atom_chain_ids:** the chain IDs corresponding to each atom in the prediction;

- **atom_plddts:** A [num_atoms] array, element i indicates the pLDDT **for atom** i in the prediction;

- **contact_probs:** A square [num_tokens, num_tokens] array. Element (i, j) indicates the predicted probability that **token i** and **token** j are in contact;

- **pae:** A square [num_tokens, num_tokens] array. Element (i, j) indicates the PAE in the position of **token** j, when the prediction is aligned to the ground truth using the frame of **token** I;

- **token_chain_ids:** A [num_tokens] array indicating the chain IDs corresponding to each token in the prediction;

- **token_res_ids:** A [num_res] array.

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AlphaFold Server tutorial Exercise 1

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Protein-ligand complex with Boltz-2 (via Neurosnap) Exercise 2

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